

Track 2F Offset-Aware Probe Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Track 2F` offset-aware probe campaign. The executed package trained the first learned `sequential_residual_offset_probe` candidate across the three required direction surfaces: `global`, `Fw`, and `Bw`.

The campaign completed all 3 planned runs with zero training failures. A terminal-level `conda run` message appeared after the runner printed the completed-campaign summary, but the generated campaign artifacts show `0` failed runs, per-run process return code `0`, and no traceback in the run logs.

The scalar leaderboard ranks `Fw` first by `test_mae`, but that ranking is only a diagnostic ordering. `Track 2F` keeps three parallel best branches for future verification and deployment analysis: one `global`, one `Fw`, and one `Bw` candidate.

Campaign Artifacts

Artifact	Path
Campaign output	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03
Campaign leaderboard	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03/campaign_leaderboard.yaml
Branch scalar pointer	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03/campaign_best_run.yaml
Execution report	output/training_campaigns/2026-06-04-11-36-09_track2f_offset_aware_probe_campaign_2026_06_03/campaign_execution_report.md
Planning report	doc/reports/campaign_plans/track2/2026-06-03-17-25-37_track2f_offset_aware_probe_campaign_plan_report.md
Campaign technical document	doc/technical/2026-06/2026-06-03/2026-06-03-17-25-37_track2f_offset_aware_probe_campaign.md
Model technical document	doc/technical/2026-06/2026-06-03/2026-06-03-18-18-20_track2f_sequential_residual_offset_probe.md
Closeout technical document	doc/technical/2026-06/2026-06-04/2026-06-04-12-28-46_track2f_campaign_closeout.md

Execution Summary

Field	Value
Campaign name	track2f_offset_aware_probe_campaign_2026_06_03
Started at	2026-06-04T11:36:09
Finished at	2026-06-04T12:04:47
Completed runs	3
Failed runs	0
Tested model type	sequential_residual_offset_probe
Tested direction scopes	global , Fw , Bw
Runner-level scalar first entry	te_sequential_residual_offset_probe_remote_fw

Directional Branch Results

Surface	Role	Run	Family	Test MAE	Test RMSE	Val MAE	Params
global	bidirectional	te_sequential_residual_offset_probe_remote_global	sequential_residual_offset_probe	0.003537	0.004005	0.003783	92,802
Fw	forward only	te_sequential_residual_offset_probe_remote_fw	sequential_residual_offset_probe_fw	0.003385	0.003931	0.003380	92,802
Bw	backward only	te_sequential_residual_offset_probe_remote_bw	sequential_residual_offset_probe_bw	0.003638	0.004280	0.003840	92,802

These three rows are the closeout result. The Fw row is the scalar first entry in this small campaign, but it does not replace the required global or Bw branch candidates.

Execution Details

Surface	Run	Status	Duration	Return Code
global	te_sequential_residual_offset_probe_remote_global	completed	9m 22s	0
Fw	te_sequential_residual_offset_probe_remote_fw	completed	12m 09s	0
Bw	te_sequential_residual_offset_probe_remote_bw	completed	7m 07s	0

Scalar Leaderboard

Rank	Surface	Run	Family	Test MAE	Test RMSE	Val MAE	Params
1	Fw	te_sequential_residual_offset_probe_remote_fw	sequential_residual_offset_probe_fw	0.003385	0.003931	0.003380	92,802
2	global	te_sequential_residual_offset_probe_remote_global	sequential_residual_offset_probe	0.003537	0.004005	0.003783	92,802
3	Bw	te_sequential_residual_offset_probe_remote_bw	sequential_residual_offset_probe_bw	0.003638	0.004280	0.003840	92,802

Technical Interpretation

The sequential residual-offset probe is execution-valid and provides the first learned branch that directly targets the mean-offset failure mode found by the `Track 2` mean-centered diagnostic, `Track 2D`, and `Track 2E`.

The scalar test metrics do not beat the current program-level scalar training winner, `te_periodic_gru_sequence_remote_Bw`, which remains at test MAE `0.002344`. This is expected for a narrow offset-aware probe: the campaign was designed to test whether the offset branch is feasible, not to promote a new global scalar winner from pointwise training alone.

The important closeout result is therefore structural. The repository now has three trained sequential residual-offset candidates that can be evaluated in the official curve-first `Track 2` surface:

- `global` : bidirectional/general candidate.
- `Fw` : forward-only candidate.
- `Bw` : backward-only candidate.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
<code>sequential_residual_offset_probe</code>	<code>te_sequential_residual_offset_probe_remote_global</code>	0.003537	New <code>global</code> <code>Track 2F</code> branch candidate
<code>sequential_residual_offset_probe_fw</code>	<code>te_sequential_residual_offset_probe_remote_fw</code>	0.003385	New forward-only <code>Track 2F</code> branch candidate
<code>sequential_residual_offset_probe_bw</code>	<code>te_sequential_residual_offset_probe_remote_bw</code>	0.003638	New backward-only <code>Track 2F</code> branch candidate

The campaign runner refreshed the family registries for all three branch families and updated the program registry. The program-level scalar best did not move because the existing Wave 2B periodic `GRU` backward-only model is still stronger on scalar `test_mae`.

Track 2 Boundary

`Track 2` was not run as part of this closeout. Under campaign governance, the official curve-first verification remains a separate operator-approved workflow after the final campaign-results report and PDF are complete. The next verification package should evaluate all three `Track 2F` branches in parallel, not only the scalar-first `Fw` row. The goal is to measure whether the offset-aware structure improves curve following and mean-offset behavior on the matching `global`, `Fw`, and `Bw` surfaces.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all three runs completed, the leaderboard and best-run artifacts exist, and the three family registries were refreshed. From a modeling standpoint, `Track 2F` is a completed feasibility branch. It is not promoted over the current scalar program winner until a separate official `Track 2` curve-first refresh confirms a real curve-level gain.

Recommended Follow-Up

1. Keep all three Track 2F branch candidates: `global` , `Fw` , and `Bw` .
2. Do not collapse Track 2F to the scalar-first `Fw` candidate.
3. Prepare the optional Track 2 curve-first verification refresh as the next operator-launched step.